Chicago Land Use Explorer

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Abstract

Examining how land use has changed over time in urban areas is crucial to gaining insight into statistical and spatial city land use trends. Land use change data can be used for predictive modeling, helping planners prepare for future urban development. This information can also highlight health and social equity concerns, allowing for targeted sustainability and environmental justice initiatives. Land use change analysis can also be used to assess the success of policy initiatives aimed at increasing or decreasing certain land uses such as green space and vacant land respectively. Data accessibility of land use change analysis can equip residents and planners with the information necessary to understand and advocate for or against such policy initiatives. Many existing land use change studies focus on the shift from natural to built environments, utilizing Landsat aerial imagery and supervised classification remote sensing algorithms for their methodology. City zoning maps are similar to urban land use data and often display current zoning visualizations online and in a widely accessible format. However, city zoning maps lack nuance for land uses such as vacancy and lack comparative analysis of previous zoning codes. This project seeks to examine, visualize, and make publicly accessible the change over time between land use category groups in Chicago (Single Family Residential, Multi-Family Residential, Commercial, Urban Mix with Residential Component, Institutional, Industrial, Transportation/Communication/Utilities/Waste (TCUW), Agricultural, Open Space, Vacant, Under Construction, Water, and Non-Parcel/Other). The Chicago Metropolitan Agency for Planning (CMAP) has created publicly available land use inventory data sets for the years 1990, 2001, 2005, 2010, 2015, 2020, and 2023. Utilizing this data along with the City of Chicago Community Areas data sets, an interactive web app was created. This web app allows planners and residents alike to create land use data visualizations by selecting between land use data

types, normalization, category groups, years, and geographic scales. Efforts to improve data accessibility for planners and the public alike through projects such as this are crucial for the future of sustainable urban development.

Introduction and Literature Review

The use of data in urban development has been contentious historically for its role in increasing inequity instead of fighting against it. Data such as the Home Owners Loan Corporation (HOLC) maps and zoning laws institutionalized redlining and systemic discrimination (Kollar et al., 2023). As a result, cities like Chicago have been invested in and built up unevenly. Socioeconomic shifts over the past 100 years have additionally created pockets of disinvestment and population loss. To combat this legacy of inequity, urban development initiatives have been implemented to improve equity and investment across Chicago.

One such initiative is the Green Healthy Neighborhoods plan. Adopted by the Chicago Plan Commission in 2014, this is a 10 to 20 year initiative that aims to revitalize vacant land in Chicago's Englewood, West Englewood, Washington Park, Woodlawn, New City, Fuller Park and Greater Grand Crossing community areas. The plan seeks to support urban agriculture, recreation, new industrial activity, and housing preservation. The City of Chicago notes that, "In 2010, the area's population of 148,000 people was less than 40 percent of its all-time high in 1940", which has led to disinvestment and a large number of vacant parcels (City of Chicago). This plan seeks to utilize these parcels to increase green space, urban agriculture, and commercial revitalization.

The City of Chicago also has a website, ChiBlockBuilder, that is associated with numerous urban planning initiatives to develop vacant parcels. These initiatives include the Missing Middle Housing Initiative which sells clusters of vacant parcels to qualified developers for $1 to build two- to six- flat apartment buildings with up to $150,000 in construction assistance available per unit (City of Chicago). ChiBlockBuilder also has an Urban Agriculture initiative that allows an applicant to purchase vacant parcels for $1 with an associated ten year commitment to use the land for urban agriculture (City of Chicago). Chicago passed a zoning ordinance in 2011 to expand the size limit on community gardens, relax fencing and parking requirements, and permit hydroponic and aquaponic systems and keeping honeybees under set conditions (City of Chicago, 2011). This zoning ordinance is the basis for ChiBlockBuilder's and other urban agriculture initiatives in Chicago.

Data analysis can be employed to evaluate the success and resulting land use change impacts of these initiatives, utilizing data to improve equity. Success from the Green Healthy Neighborhood plan could be quantified by reduction of vacant parcels and an increase of open space, residential, industrial, and commercial parcels in the years since plan implementation in 2014. Success from the 2011 zoning ordinance and ChiBlockBuilder's Urban Agriculture initiative could be quantified similarly with a reduction of vacant parcels and increase of open space and agriculture parcels. The Missing Middle Housing Initiative is newer, having first launched in Fall 2024, so it will take time to see associated land use change resulting from this program (Miller, 2025). Land use change data analysis and accessibility provide planners and residents alike with the tools to evaluate the success of urban development initiatives like Green Healthy Neighborhoods and ChiBlockBuilder and work together to improve land use equity in Chicago.

A significant portion of the existing literature on land use change focuses on change from natural land uses (grassland, bare land, agriculture, etc.) to built environment and seeks to quantify urban development and sprawl. A study by Wang et al. (2018) quantified the land use/land cover change (LULCC) for the city of Dongying, China, between 1995 and 2015. They found a reduction across grassland, farmland, water, and bare land, with a corresponding 347.34% increase in built-up land. Addae and Oppelt (2019) found similar patterns in their LULCC study of the Greater Accra Metropolitan Area (GAMA), Ghana, reporting a 277% increase in built environment between 1991 and 2015. Forested areas in this study suffered the greatest decrease in land coverage from 34% land cover in 1991 to just 6.5% in 2015. Hassan et al. (2016) reported a 213.5% increase in built land between 1992 and 2012 in their study on Islamabad, Pakistan, with a 49.42% decrease in forest area and 96.63% decrease in barren land over the same period.

The methods employed by these land use change analyses are also significantly different from those employed by this study. Many land use change analyses utilize Landsat satellite imagery and supervised classification remote sensing algorithms to determine land categories and assess changes. Supervised classification involves a user setting established classification groups and providing the model with training samples of pre-identified Landsat imagery pixels to learn the classification schema so it can be applied to a larger dataset (ESRI, n.d.). The Wang et. al (2018) study in Dongying, China, the Addae and Oppelt (2019) study of the Greater Accra Metropolitan Area (GAMA), Ghana, and the Hassan et al. (2016) Islamabad, Pakistan study all utilized Landsat satellite imagery as the data input for their supervised classification maps and models. This methodology is successful for quantifying change across visually distinct land use types (grassland to built environment) but is not an ideal approach in discerning between utilization changes in built environments (a building use changing from institutional to commercial).

Another approach utilized to quantify urban land use are zoning maps created by local governments and other actors. The City of Chicago produces and hosts its own zoning web maps on the City of Chicago website. Second City Zoning is another Chicago zoning web map created with open source code and data and is maintained by a civic technology company, DataMade (Second City Zoning). Most large metropolitan areas (New York, Los Angeles, Phoenix) host interactive web maps on their websites while smaller localities may have static zoning pdf maps available (NYC Department of City Planning; Los Angeles City Planning; City of Phoenix). Zoning code maps miss the nuance identified at the parcel level for land use types not typically included within zoning codes. For example, vacancy is not a zoning code but is a crucial piece of data when considering historical trends and making future policy decisions. Additionally, like most of the zoning web maps the City of Chicago and Second City Zoning web maps display only current zoning information with no historical data, which lacks the capacity for change and historical trend analysis. Some cities such as New York maintain historical zoning data files on their website but do not actively visualize them or offer comparison analysis between years.

There is a gap in the literature seeking to analyze how an existing urban environment has shifted through time. The input data utilized and scope of this research is novel. The Chicago Metropolitan Agency for Planning's Land Use inventory data is unlike other land use datasets as the data offers unique insights not captured by zoning maps or remote sensing analysis of Landsat imagery. This project provides accessible and user-friendly Chicago land use change insights for planners and residents alike.

Methods

The Chicago Metropolitan Agency for Planning (CMAP) has created publicly available land use inventory data sets for the years 1990, 2001, 2005, 2010, 2015, 2020, and 2023 (CMAP, 2022; CMAP, 2022; CMAP, 2022; CMAP, 2016; CMAP, 2022; CMAP, 2024; CMAP, 2026). This data indicates the land use category for each land parcel in the seven Northeastern Illinois counties (Cook, DuPage, Kane, Kendall, Lake, McHenry, and Will counties). Utilizing this data along with the City of Chicago Community Areas data, an interactive web app was created (City of Chicago, 2026). This web app allows planners and residents alike to create various land use spatial and statistical visualizations using drop down menus to select between data types, normalization, years, and land use category groups.

The land use inventory data sets for each year were downloaded as geodatabase files along with the Chicago Community Area shapefile and opened within ArcGIS Pro for initial processing. Each year of the land use inventory data sets were re-projected as necessary to NAD 1983 State Plane Illinois East FIPS 1201 (US Feet). This was completed so that the coordinate projections match across all of the data and that projection distortion would be minimized for later geometry calculations. The correctly projected land use inventory data were clipped to the Chicago boundaries, and spatial joins were used to assign community area data to each parcel within Chicago boundaries. Manual spatial join post processing was completed as each year had between 25-45 parcels remaining with a null Community Area assignment due to the spatial join match option being set to ‘has its center in’. The geometry for each parcel was calculated in US Survey Square Feet in a new field labelled ‘Area’ for each year of data. Summary statistics were employed to summarize the parcel area and parcel count of each land use category within each community area for all seven data years. The summary statistics tables were pivoted wider by

Land Use twice each to create one pivot table for area values and a second pivot table for parcel count values for each data year. These 14 pivot tables were each rejoined to an individual Community Area shapefile, which produced 14 feature classes with land use categories as the columns, community areas as the rows, and values indicating either summed parcel area or parcel counts for each land use category in each community area. The 14 feature classes were reprojected to the WGS 1984 coordinate reference system and exported as GeoJSON files for further analysis and creation of an interactive web app using Streamlit in Visual Studio Code (Streamlit, 2026; Microsoft, 2026).

There are 57 unique land use categories in Chicago within the most recent year of the raw data. The unique land use categories have evolved since the initial land use data set created in 1990 and do not follow the same categorization schema for every year. As such, a universal schema was created using python in Visual Studio code to aggregate unique land use categories from all years of data into 13 new land use category grouping fields inspired by the ones used on CMAP's Land Use Inventory web map (Appendix A). These 13 land use category groupings include: Single Family Residential, Multi-Family Residential, Commercial, Urban Mix with Residential Component, Institutional, Industrial, Transportation/ Communication/ Utilities/Waste (TCUW), Agricultural, Open Space, Vacant, Under Construction, Water, and Non-Parcel/Other. The aggregated values for the 13 land use category grouping fields in the seven parcel area GeoJSONs were converted from square feet to square miles.

A Chicago Land Use Explorer web app was created with Streamlit, Pandas, NumPy, Matplotlib, Altair, Folium, Streamlit-Folium, GeoPandas, Shapely, and Seaborn python libraries in Visual Studio code with aid from the built-in AI code editor and Microsoft Copilot. The web app was deployed using the Streamlit python framework and features three interactive subpages:

Map, Charts, and Change Analysis. The Map subpage allows users to create land use percentage maps by utilizing drop downs to select the data type (parcel area vs parcel count), year of data (1990, 2001, 2005, 2010, 2015, 2020, or 2023) and land use category grouping (Single Family Residential, Multi-Family Residential, Commercial, Urban Mix with Residential Component, Institutional, Industrial, Transportation / Communication / Utilities / Waste (TCUW), Agricultural, Open Space, Vacant, Under Construction, Water, and Non-Parcel / Other.). The Charts subpage allows users to select data type, normalization (Percentage, Square Miles, or Count depending on selected data type), data years and community areas to produce various statistical charts. Histograms and pie charts compare the normalization breakdown of each land use category grouping for the selected year and geography. Users can additionally select specific land use category groups to create multi-year trend line charts for the user selected data. The Change Analysis subpage allows users to create diverging graduated color maps displaying the percentage change by selecting data type, normalization, two unique years of land use data, and a land use category group.

Results and Conclusion

The Chicago Land Use Explorer web app can be utilized to evaluate the impact of the Green Healthy Neighborhood plan in Chicago's Englewood, West Englewood, Washington Park, Woodlawn, New City, Fuller Park and Greater Grand Crossing community areas. A percentage change matrix was assembled by creating change analysis maps for each of the relevant land use category groups (vacant, open space, commercial, industrial, single family residential, and multi-family residential) for parcel area (square miles) between 2015 and 2023. This matrix can be seen in Figure 1. Vacant land use saw a 6.71% decrease in Englewood, 13.36% increase in West Englewood, 13.99% decrease in Washington Park, 14.14% decrease in Woodlawn, 4.37%

increase in New City, 26.52% increase in Fuller Park, and a 6.67% decrease in Greater Grand Crossing (Figure 1). This indicates mixed results of vacant land reduction.

Recreation open space, conservation open space, and community gardens or urban agriculture all fall into the Open Space land use category group. Open space parcel area saw a 3.30% increase in Englewood, 2.56% increase in West Englewood, 0.02% decrease in Washington Park, 6.16% decrease in Woodlawn, 3.94% increase in New City, 14.73% increase in Fuller Park, and a 3.83% increase in Greater Grand Crossing (Figure 1). Though there are certainly other confounding variables, it is likely that the Green Healthy Neighborhood plan had a generally positive impact on open space in the targeted community areas.

Single family residential parcel area percent change revealed an increase in 3 community areas and a decrease in the other 4, indicating mixed results (Figure 1). Washington park notably saw an 85.77% increase in commercial area and Woodlawn saw a 26.81% increase in industrial area. Multi-family residential parcel area percent change reveals a small increase in 2 community areas and a decrease in the other 5, indicating this policy may not have helped to preserve multi-family residential housing (Figure 1).

	Vacant	Open Space	Commercial	Industrial	SF Residential	MF Residential
Englewood	-6.71%	+3.30%	+11.87%	+18.67%	-6.33%	-4.51%
West Englewood	+13.36%	+2.56%	-2.49%	+0.29%	-5.07%	-6.47%
Washington Park	-13.99%	-0.02%	+85.77	-7.10%	+3.43%	+1.65%
Woodlawn	-14.14%	-6.61%	-1.31%	+26.81%	+10.21%	+1.31%
New City	+4.37%	+3.94%	+5.09%	+0.14%	+1.08	-3.07%
Fuller Park	+26.52%	+14.76%	+1.07	-38.60%	-2.22%	-4.12%
Greater Grand Crossing	-6.67%	+3.83%	-6.19%	-8.09%	-1.05%	-1.13%

Figure 1. Green Healthy Neighborhoods Plan Impact Matrix (Positive change indicated in white)

The Chart page can provide insight on how open space has changed in all of Chicago across the seven data years. The multi-year trend chart on the Chart page reveals that open space in Chicago has not significantly changed between 1990 and 2023; open space has hovered around just above 16 square miles or 7% of Chicago's total land use across all years of data, seen in Figure 2. This indicates that the 2011 zoning ordinance and Urban Agriculture initiatives have not caused a significant increase in open space square mileage for the city as a whole.

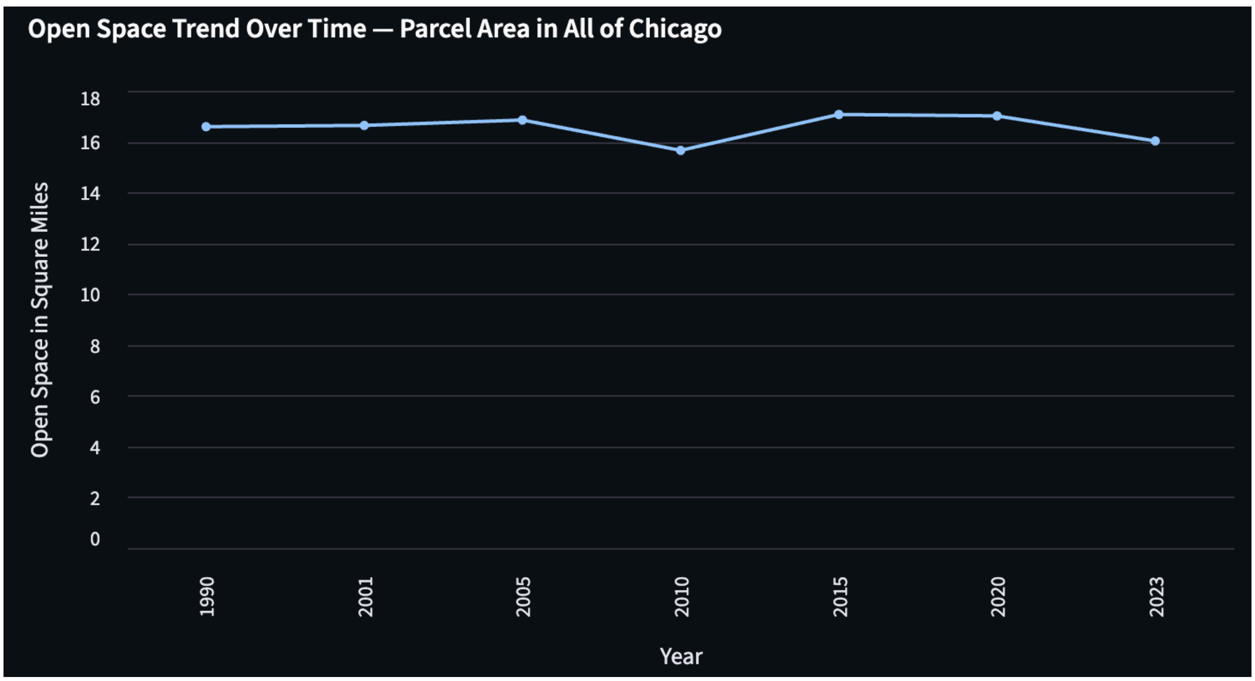

Figure 2. Open Space (square miles) in all of Chicago 1990-2023

Englewood saw mixed impacts including beneficial changes to vacant, open space, commercial, and industrial land use categories between 2015 and 2020 possibly in relation to the Green Healthy Neighborhood plan and Urban Agriculture initiatives, but dramatic inequities remain. Creating a percentage map of vacancy percentage across Chicago in 2023 reveals that Englewood has the second highest vacancy rate of all Chicago community areas at 21.1% (Figure 3). Englewood focused community organizations like the Resident Association of

Greater Englewood (R.A.G.E.) could utilize this data in awareness and fundraising initiatives to motivate residents to demand further change in their community.

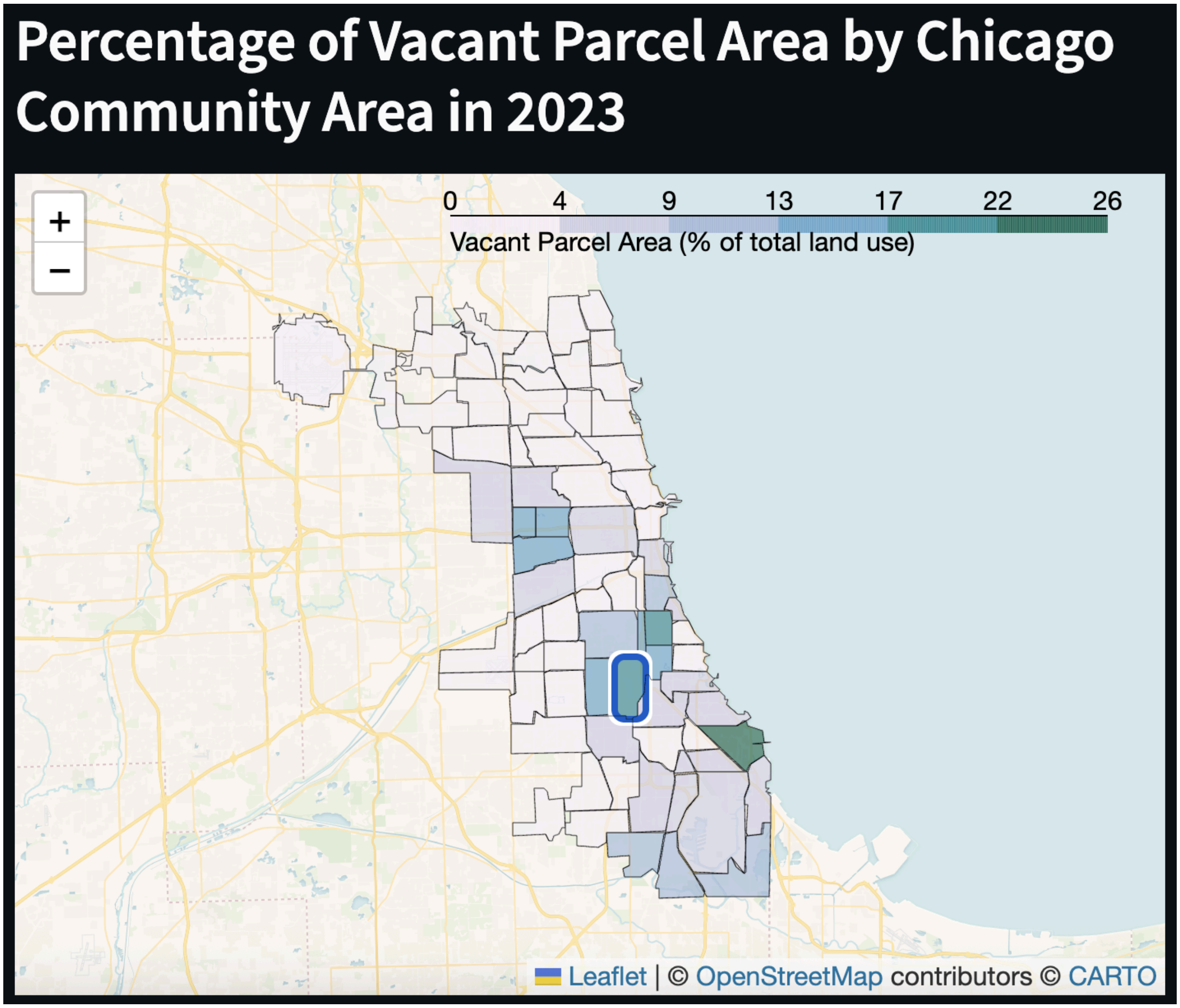

Figure 3. Graduated color percentage map of vacancy rate (square miles) in Chicago by community area. Englewood is highlighted with a blue and white border

An important data limitation relevant for interested parties using the Chicago Land Use Explorer web app to create visualizations is the consideration of data structure changes across the years of CMAP data. CMAP's 1990 dataset has just under 19,000 parcels in Chicago and approximately 177,000 Chicago parcels in the 2023 dataset. Additionally, the land use categories in the earlier data years were different than the later data years. This means that some of the observed changes in land use category groups over time may be due to changes in data production methods rather than actual land use changes on the ground. CMAP's land use categories have been standardized since 2010, so the most accurate comparisons can be made between 2010, 2015, 2020, and 2023.

The Chicago Land Use Explorer web app expands land use data and trend accessibility for residents and planners, highlighting modular insights based on the user's input. It is a useful tool for understanding spatial and statistical trends of land use change in Chicago between 1990 and 2023.

Appendices

Appendix 1. Land use category universal schema matrix 1990-2023

Land Use Category	1990	2001	2005	2010	2015, 2020, and 2023
Single Family	1110, 1120, 1140	1110, 1120, 1140	1110, 1120, 1140	1111, 1112, 1140	1111, 1112, 1140
Multi-Family	1130	1130	1130	1130	1130
Commercial	1210, 1220, 1230, 1241, 1242, 1243, 1250, 1260	1211, 1212, 1221, 1222, 1223, 1231, 1232, 1240, 1250	1211, 1212, 1221, 1222, 1223, 1224, 1231, 1232, 1240, 1250	1211, 1212, 1214, 1215, 1220, 1240, 1250	1211, 1212, 1214, 1215, 1220, 1240, 1250
Urban Mix + Residential	N/A	N/A	N/A	1216	1216
Institutional	1311, 1312, 1313, 1320, 1330, 1340, 1350, 1360, 1370, 1380, 1390	1310, 1320, 1330, 1340, 1350, 1360, 1370	1310, 1320, 1330, 1340, 1350, 1360, 1370	1310, 1321, 1322, 1330, 1340, 1350, 1360, 1370, 1380	1310, 1321, 1322, 1330, 1340, 1350, 1360, 1370, 1380
Industrial	1410, 1420, 1430, 1440	1410, 1420, 1430, 1440	1410, 1420, 1430, 1440	1410, 1420, 1431, 1432, 1433, 1450	1410, 1420, 1431, 1432, 1433, 1450
TCUW	1510, 1520, 1530, 1540, 1550, 1560	1511, 1512, 1520, 1530, 1540, 1550, 1560	1511, 1512, 1520, 1530, 1540, 1550, 1560	1511, 1512, 1520, 1530, 1540, 1550, 1561, 1562, 1563, 1564, 1565, 1570	1511, 1512, 1520, 1530, 1540, 1550, 1561, 1562, 1563, 1564, 1565, 1570
Agricultural	2000	2100, 2200, 2300	2100, 2200, 2300, 2400	2000	2000
Open Space	3110, 3120, 3130, 3210, 3220	3100, 3200, 3300, 3400, 3500, 3600	3100, 3200, 3300, 3400, 3500, 3600	1151, 3100, 3200, 3300, 3400, 3500, 6100	1151, 3100, 3200, 3300, 3400, 3500
Vacant	4110, 4120, 4130, 4300	4110, 4120, 4300	4110, 4120, 4300	4110, 4120, 4130, 4140	4110, 4120, 4130, 4140
Under Construction	4210, 4220	4210, 4220	4210, 4220	4210, 4220, 4230, 4240	4210, 4220, 4230, 4240
Water	5100, 5200, 5300	5100, 5200, 5300	5100, 5200, 5300	5000, 6200	5000
Non-Parcel / Unclassified	9999	9999	9999	6300, 6400, 9999	6000, 9999

Appendix 2. Web app link address

https://chicago-landuse-app.streamlit.app/

Works Cited

Addae, B., & Oppelt, N. (2019). Land-use/land-cover change analysis and Urban Growth Modelling in the Greater Accra Metropolitan Area (gama), Ghana. *Urban Science*, *3*(1), 26. https://doi.org/10.3390/urbansci3010026

City of Chicago. (2011). *Mayor Emanuel introduces an ordinance supporting Urban Farms*. City of Chicago :: Mayor Emanuel Introduces An Ordinance Supporting Urban Farms. https://www.chicago.gov/city/en/depts/dcd/provdrs/planning_and_policydivision/news/2011/aug/mayor_emanuel_introducesanordinancesupportingurbanfarms.html

City of Chicago (2026). *Boundaries - Community Areas* [Data set]. City of Chicago. https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Community-Areas/igwz-8jzy/about_data

City of Chicago. (n.d.-a). *City of Chicago: Zoning Website*. https://gisapps.chicago.gov/ZoningMapWeb/?liab=1&config=zoning

City of Chicago. (n.d.-b). *Green Healthy Neighborhoods*. City of Chicago :: Green Healthy Neighborhoods. https://www.chicago.gov/city/en/depts/dcd/supp_info/green-healthy-neighborhoods.html

City of Chicago. (n.d.-c). *Missing middle infill housing: Missing middle infill housing*. Missing Middle Infill Housing | Missing Middle Infill Housing. https://www.chicago.gov/city/en/sites/missing-middle/home.html

City of Chicago. (n.d.-d). Urban Agriculture. https://www.chicago.gov/city/en/sites/block-builder/home/application-guide/urban-agriculture.html

City of Phoenix. (n.d.). *My community map*. https://www.phoenix.gov/administration/departments/pdd/tools-resources/maps/my-community-map.html

CMAP (2016). *Chicago Land Use Inventory (2010)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/datasets/CMAPGIS::land-use-inventory-northeastern-illinois-2010/about

CMAP (2022). *Chicago Land Use Inventory (2005)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/datasets/CMAPGIS::land-use-inventory-northeastern-illinois-2005/about

CMAP (2022). *Chicago Land Use Inventory (2001)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/datasets/CMAPGIS::land-use-inventory-northeastern-illinois-2001/about

CMAP (2022). *Chicago Land Use Inventory (1990)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/datasets/CMAPGIS::land-use-inventory-northeastern-illinois-1990/about

CMAP (2022). *Chicago Land Use Inventory (2015)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/datasets/CMAPGIS::land-use-inventory-northeastern-illinois-2015/about

CMAP (2024). *Chicago Land Use Inventory (2020)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/maps/c7959bd1c0084edba3264099deeaf365/about

CMAP (2026). *Chicago Land Use Inventory (2023)* [Data set]. Chicago Metropolitan Agency for Planning. https://datahub.cmap.illinois.gov/maps/1022fd982a5b4b51a9cecb20ea6c2a29/about

ESRI. (n.d.). *Overview of Image Classification*. Overview of image classification-ArcGIS Pro | Documentation. https://pro.arcgis.com/en/pro-app/latest/help/analysis/image-analyst/overview-of-image-classification.htm

Hassan, Z., Shabbir, R., Ahmad, S. S., Malik, A. H., Aziz, N., Butt, A., & Erum, S. (2016). Dynamics of Land Use and land cover change (LULCC) using geospatial techniques: A case study of islamabad pakistan. *SpringerPlus*, *5*(1). https://doi.org/10.1186/s40064-016-2414-z

Kollar, J., McGlashan, N., & Williams, S. (2023). *Creating action with data: Using data to increase equity in Urban Development | Joint Center for Housing Studies*. Harvard University Joint Center for Housing Studies.

https://www.jchs.harvard.edu/research-areas/working-papers/creating-action-data-using-data-increase-equity-urban-development

Los Angeles City Planning. (n.d.). *Zimas*. Zimas LA City. https://zimas.lacity.org/

Microsoft. (2026). *Visual Studio Code* (Version 1.90) [Computer software]. https://code.visualstudio.com/

Miller, A. (2025, May 26). *Chicago's "missing middle" housing program adds more neighborhoods, extends developer deadline*. Times. https://chicago.suntimes.com/real-estate/2025/05/26/chicago-missing-middle-housing-program-developer-deadline

NYC Department of City Planning. (n.d.). *NYC GIS Zoning Features*. Visit nyc.gov. https://www.nyc.gov/content/planning/pages/resources/datasets/gis-zoning-features

R.A.G.E. (n.d.). *Resident association of greater englewood*. Resident Association of Greater Englewood. https://ragenglewood.org/

Streamlit. (2026). *Streamlit: A faster way to build and share data apps* (Version 1.57.0) [Computer software]. https://streamlit.io/

Wang, C., Wang, Y., Wang, R., & Zheng, P. (2018). Modeling and evaluating land-use/land-cover change for urban planning and Sustainability: A case study of Dongying City, China. *Journal of Cleaner Production*, *172*, 1529–1534. https://doi.org/10.1016/j.jclepro.2017.10.294

Zoning, 2nd City. (n.d.). *2nd city zoning - an interactive map of Chicago's zoning code*. 2nd City Zoning - An interactive map of Chicago's zoning code. https://secondcityzoning.org/